HW: Labs 5 14.11.26

4.17) $P_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$

a) $X \sim Normal(\mu, \sigma)$ $\log(Y) = X \Rightarrow Y = e^X$

we know $g(x) = e^x$ is a strictly monotonic, continuously differentiable function, therefore

$$P_Y(y) = P_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right| =$$

$$= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(y)-\mu)^2}{2\sigma^2}\right) \cdot \frac{1}{y} =$$

$$= \frac{1}{y\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\log(y)-\mu)^2}{2\sigma^2}\right), y > 0$$

b) ✓

4.12) $X|\mu, \sigma^2 \sim Normal(\mu, \sigma^2)$

$$P_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

a) $Z = \frac{X-\mu}{\sigma^2} \sim N(0,1)$

$$P(Z \le z) = P\left(\frac{X-\mu}{\sigma} \le z\right) = P(X \le \mu + z\sigma^2) =$$

$$= F_X(\mu + z\sigma^2) = \int_{-\infty}^{\mu+z\sigma^2} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx =$$

let $t = \frac{x-\mu}{\sigma^2}$, then $\frac{dt}{dx} = \frac{1}{\sigma^2}$, $\sigma\, dt = dx$

$x \to -\infty : t \to -\infty$

$x \to \mu + z\sigma^2 \quad t \to \frac{\mu + z\sigma^2 - \mu}{\sigma^2} = z$

$$= \int_{-\infty}^{\mu+z\sigma} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right) dx$$

$$= \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi\sigma^2}} \sigma \exp\left(-\frac{t^2}{2}\right) dt$$

$$= \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

this is the same as CDF of $Y \sim N(0,1)$! Thus $Z \sim N(0,1)$

b) ✓ a) ✓

4.11) ↑ Modelling of positive values - rainfalls

$X|\alpha,\beta \sim \text{Gama}(\alpha,\beta)$, $\alpha>0, \beta>0$, $x \in [0,\infty)$

shape / rate

$$p_X(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

$$F_X(x) = \frac{\gamma(\alpha,\beta x)}{\Gamma(\alpha)} \quad , \quad \gamma(s,x) = \int_0^x t^{s-1} e^{-t}\,dt$$

a) $X \sim \text{Gama}(1,\beta)$

$$p_X(x) = \frac{\beta}{\underbrace{\Gamma(1)}_{0!=1}} \cdot 1 \cdot e^{-\beta x} = \underline{\underline{\beta e^{-\beta x}}}$$

$X \sim$ Exponential (β) !

b) $k=\alpha$, $\theta = \frac{1}{\beta}$ $\quad X|k,\theta \sim \text{Gama}(k,\theta)$

$\beta = \frac{1}{\theta}$ ←

$$p_X(x) = \frac{1}{\theta^k \Gamma(k)} x^{k-1} e^{-\frac{x}{\theta}}$$

c) ✓

4.10) $X|\alpha,\beta \sim \text{Beta}(\alpha,\beta)$

$$p_X(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)} \qquad B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$\Gamma(z) = \int_0^\infty x^{z-1} e^{-x}\,dx$$

suport en $[0,1]$

$\alpha,\beta > 0$

b) ✓

c) ✓

a) $\alpha=\beta=1$

$$p_X(x) = \frac{1}{B(1,1)} = \frac{1}{\frac{\Gamma(1)\Gamma(1)}{\Gamma(2)}} = \frac{1}{\frac{1}{1}} = \underline{\underline{1}}$$

Uniform $(0,1)$ distribution!